

Continuous $\times$ continuous interactions in linear regression

This crib sheet covers interactions between continuous predictor variables in multiple regression analysis.

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The theory

An interaction occurs when the combined effect of two predictors on the outcome variable is different than just the sum of their separate effects.

There are three main possibilities when predictors X and Z interact in predicting Y :

1. Additive effects – The effect equals the sum of X 's effect and Z 's effect separately.
No interaction.
2. Synergistic interaction – The combined effect is greater than the sum of the parts.
Having both high X and high Z leads to much higher Y than expected from just adding their separate effects.
3. Compensatory interaction – The predictors trade-off or compensate. For very high X , increases in Z matter less for predicting Y . And vice versa.

When there is an interaction, the regression of Y on one predictor (e.g. X) depends on the value of the other predictor (Z). The effect of X on Y changes as Z changes.

Interactions are tested by adding a product term (XZ) to the regression equation along with the original X and Z predictors. The product term represents the joint effect above and beyond the additive effects.

Cohen et al. (2003) given a demonstration of plotting such interactions. Plotting interactions show how the increments in Y depend on both X and Z jointly through their product XZ , creating a curved 3D surface, rather than just a flat additive plane.

You might be wondering why we would want to bother with this procedure. Lots of statistics textbooks suggest something a little simpler in these circumstances. They suggest dichotomising continuous variables and then using an ANOVA procedure. Because almost all software implementations of ANOVA include testing an interaction effect automatically, this seems easier. But Cohen et al. (2003) warn strongly against this. They point out that the reliance on ANOVA is partly just because the methods for testing out interactions in ANOVA were developed first, and people had got used to relying on this approach a while before the regression approach emerged.

Most textbooks do mention that dichotomisation of a continuous variable leads to a loss of power. But Maxwell & Delaney (1993) demonstrate that the dichotomisation-and-ANOVA method can actually lead to false positive results in some circumstances too. In short: don't do it.

Example

Let's say we are trying to predict job performance (Y) from measures of cognitive ability (X) and motivation (Z) in a sample of employees.

An additive model with no interaction would be: $Y = 0.5X + 0.4Z + 5$

This means that for every 1 unit increase in cognitive ability, job performance goes up by 0.5 units. And for every 1 unit increase in motivation, job performance goes up by 0.4 units. The effects are simply additive.

However, we may hypothesise a synergistic interaction such that employees high on both ability and motivation perform much better than expected from just adding the separate effects. The regression equation would end up taking a form such as: $Y = 0.5X + 0.4Z + 0.3XZ + 3$

The XZ product term represents the interaction effect. When both X and Z are high, their product is large, providing a synergistic boost to job performance above and beyond the additive effects.

Alternatively, a compensatory interaction could occur if high ability can compensate for low motivation, and vice versa. Then the equation may be: $Y = 0.5X + 0.4Z - 0.2XZ + 5$

Now when X and Z are both high, making XZ large, it subtracts from predicted performance rather than adding synergistically.

Plotting these interactions would show different curved 3D surfaces, either rising up sharply for synergistic effects when X and Z are both high, or flattening out for compensatory effects. More on that later.

Five Step Process

The process for testing continuous $\times$ continuous interactions popularised by Cohen and colleagues can be summarised thus:

1. Center the Predictor Variables
2. Construct Interaction Term by multiplying the two centred variables
3. Run Regression Analysis
 - Request unstandardized predicted values to be saved
4. Interpret Coefficients
 - Remember, we cannot interpret coefficients independently due to the interaction
 - Interpret as: Influence of X on Y depends on level of Z
 - Significant interaction term means the variables depend on each other for their effect on the predicted variable
5. Plot the Interaction
 - Create buckets/groups for one of the predictor variables based on ± 1 SD
 - Do a scatter plot with separate markers for each bucket of the predictor that was grouped just above
 - Add regression fit lines for each bucket to visualise changing slopes
 - Interpret how slopes differ across levels of moderator

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References

Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2003). *Applied multiple correlation/regression analysis for the behavioral sciences*. Lawrence Erlbaum Associates.

Maxwell, S. E., & Delaney, H. D. (1993). Bivariate median splits and spurious statistical significance. *Psychological bulletin*, 113(1), 181.